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C+ KioskEcpi – Integration

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3.3	2024-05-03	SF Lee	Added: - Error code for parking
3.4	2024-05-16	SF Lee	Added: - To support TNG card (Init Entry / Init Entry Status / Proceed Entry / Init Exit / Init Exit Status / Proceed Exit) - Proceed Exit Status - SetBlacklist
3.5	2024-07-12	SF Lee	Added: - Process Flow for Parking Entry and Exit - payFlag for parking
3.6	2024-07-26	SF Lee	Updated: - Process Flow Fixed: - Listening port 5000
3.7	2024-10-08	Justin Aw	Added: - The recommended logging from kiosk. - Process Flow State Diagram. - Signature calculation and Signed Message.
3.8	2024-10-24	Justin Aw	Added: - setAutoReloadFlag for TNG
3.9	2024-11-26	Seow Yen	Added - Set reader IP using TCPI tool Updated - Process Flow

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1 Introduction

1.1 Scope

This document outlines the integration API between reader and controller:

1. Overview, Connectivity & Setup
2. Message format
3. Commands & Responses
4. Process Flow

1.2 Abbreviations

EMV	Europay, Mastercard, Visa
Jellies	CoherentPlus Contactless Reader
ODA	Offline Data Authentication
Kiosk	Kiosk Controller
LUS	Lane Unattended Server
TNG	Touch N Go

1.3 References

NA

2 Overview, Connectivity & Setup

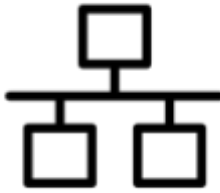
2.1 Overview



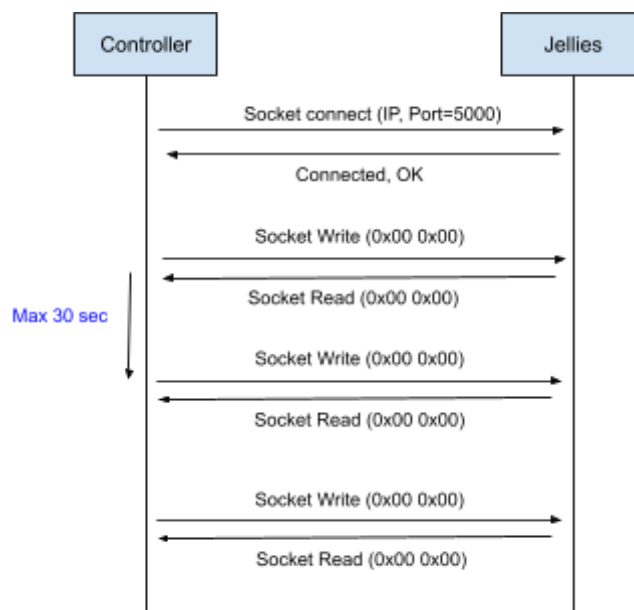
ECPI is a communication interface that facilitates payment functionality for kiosks to accept EMV card, local electronics purse payment and E-Wallet. Kiosk and Jellies reader are connected via LAN communication. Jellies reader and LUS are connected via the internet. This document describes the command to operate Jellies to accept payment.

The commands are designed for simple integration, without being required to integrate the bank system. Jellies will handle the EMV, local electronics purse card reading, E-Wallet and bank host authorization process, and returning kiosk controller a payment status with approval detail.

2.2 Connectivity & Setup



- A. Socket (TCP via LAN)
 - a. Default socket is opened at Port 5000
 - b. Connect to this device IP with port number
 - c. **One connection allowed per session. Second connection will be refused if the first connection is still alive.**
- B. Heartbeat
 - a. **Heartbeat is required to send in every 30 seconds (within 30 seconds) if there is no communication in between.**
 - b. Connection will be dropped by the reader if there is no heartbeat communicated within the heartbeat cutoff period. When the connection is dropped, the controller is required to recover the connection by reestablishing the connection and initialization process as per illustrated in [Process Flow](#).
 - c. Heartbeat is a 2 bytes message carrying null bytes, 0x00 0x00.



3 Message Format

This chapter describes the general format of requests and responses that communicate between reader and kiosk. The supported message format for all the requests and responses is in JSON format.

3.1 Request

Key Name	Max Length	Description
apiVersion	12	Api Version.
message	20	Command of this message.
type	10	Type of this message. request - Request
timestamp	20	Date time of this message. 2023-05-30 11:06:02 - YY-MM-DD hh:mm:ss
messageTraceID	40	Trace Id for this message.
body	n	Body of the message.
signature	64	SHA256 of the full message with a secret key. signature = secretkey + full message

Example:

```
{
  "apiVersion": "1.0",
  "message": "initTerminal",
  "type": "request",
  "timestamp": "2023-05-30 11:06:02",
  "messageTraceID": "112233445566",
  "body": {
    "plazaID": "P01",
    "laneID": "L01",
    "operationMode": "1",
    "laneType": "3"
  },
  "signature": "0E5962DAC59DA96552E00BF7FD0333FA7E49C9011EE465948AD92C35BE80A706"
}
```


3.2 Response

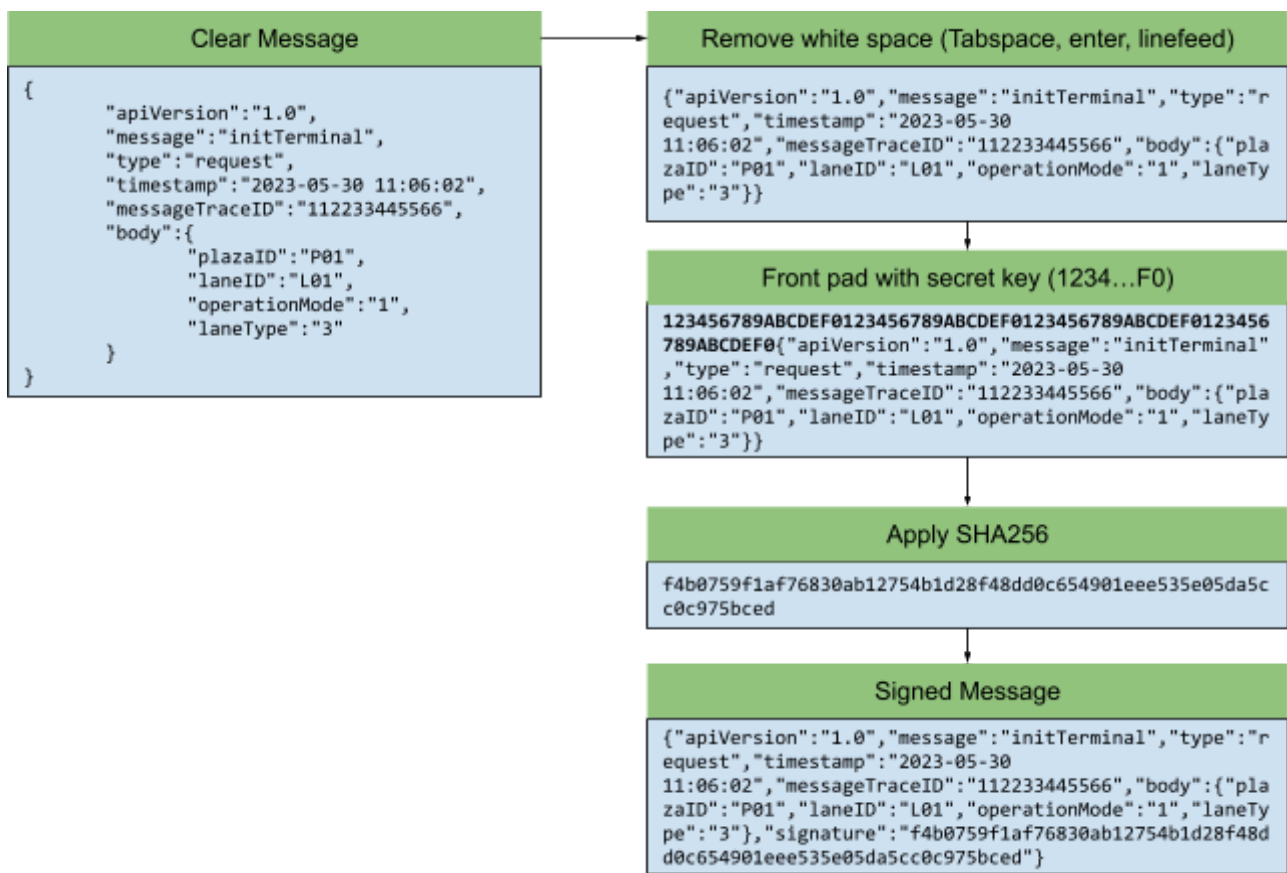
Key Name	Max Length	Description
apiVersion	12	Api Version.
message	20	Command of this message.
type	10	Type of this message. ack - Acknowledgement
timestamp	20	Date time of this message. 2023-05-30 11:06:02 - YY-MM-DD hh:mm:ss
messageTraceID	40	Trace Id for this message.
body	n	Body of the message.
		errorCode[4] Response status of command execution
signature	64	SHA256 of the full message with a secret key. signature = secretkey + full message Please refer to section Signature Calculation and Signed Message

Example:

```
{
  "apiVersion": "1.0",
  "message": "initTerminal",
  "type": "ack",
  "timestamp": "2023-05-30 11:06:02",
  "messageTraceID": "112233445566",
  "body": {
    "errorCode": "0000"
  },
  "signature": "ADE4FC34CEBCB47B9493ED1277F038E90A7DC7EC4ADB3125D2BA38314ACE60FA"
}
```

3.3 Signature Calculation and Signed Message

The flow diagram below illustrates the signature calculation and generating a signed message.



3.4 Logging between Kiosk Controller and Reader

System integrators are required to log all communications between the controller and the reader. This should include timestamps, IP addresses, network connection activities, socket messages, and any exceptions that occur during message exchanges, as shown below:

A. A normal scenario:

```
2024-05-23 10:05:10 [192.168.0.50] >> Connect Port 500
2024-05-23 10:05:10 [192.168.0.50] << Connected Successfully
2024-05-23 10:05:11 [192.168.0.50] >> SENT "{abcdefg.....}"
2024-05-23 10:05:12 [192.168.0.50] << RCVD "{abcdefg.....}"
2024-05-23 10:05:13 [192.168.0.50] << RCVD "{abcdefg.....}"
2024-05-23 10:05:14 [192.168.0.50] >> SENT "{abcdefg.....}"
2024-05-23 10:05:15 [192.168.0.50] << RCVD "{abcdefg.....}"
2024-05-23 10:05:16 [192.168.0.50] >> Disconnect 500
2024-05-23 10:05:16 [192.168.0.50] << Disconnected Successfully
```

B. Example logs of exception errors.

```
2024-05-23 10:05:10 [192.168.0.50] >> Connect Port 500
2024-05-23 10:05:10 [192.168.0.50] << ERROR: Failed to connect to port 500 (Conn refused)

2024-05-23 10:05:11 [192.168.0.50] >> SENT "{abcdefg.....}"
2024-05-23 10:05:15 [192.168.0.50] << ERROR: Failed to write on socket.

2024-05-23 10:05:11 [192.168.0.50] >> SENT "{abcdefg.....}"
2024-05-23 10:05:15 [192.168.0.50] << ERROR: Timeout. Unable to receive response.

2024-05-23 10:09:16 [192.168.0.50] << ERROR: Connection terminated.

2024-05-23 10:15:18 [192.168.0.50] << ERROR: Connection closed by remote device
```

This log is crucial for debugging and issue analysis between the system integrator and CoherentPlus. In the event of exceptional cases during development or in a live environment, the log should be submitted to CoherentPlus for initial troubleshooting and resolution.

It is mandatory to retain this log in the controller for at least 15 days, and doing so is highly advisable for effective issue tracking.

4 Commands & Responses

This section outlines the command set and its response respectively. A valid command message shall be packed with message format as described in the previous chapter.

4.1 Init Terminal

The reader will always be in "not-in-use" mode after start up. This command activates the reader into "ready to use" state. Kiosk is required to pass in the plazaId, laneId, operationMode, laneType. The timestamp will be used to sync the date time to the reader.

Request

Key Name	Max Length	Value	
apiVersion	12	1.0	
message	20	initTerminal	
type	10	request	
timestamp	20	2023-05-30 11:06:02	
messageTraceID	40	112233445566	
body	n	plazaID[10]	P01
		laneID[10]	L01
		operationMode[2]	0 - Maintenance Mode 1 - Live Mode 2 - Not In Use
		laneType[2]	1 - Entry 2 - Exit 3 - Open 4 - Dual
		Optional Fields	
		title[18]	Optional. Title text to display
		message[40]	Optional. Message text to display
		selectNext[2]	Optional. To allow unknown card to be processed by controller on APDU level 0 - Unknown card will be alert with error message 1 - Unknown card will NOT be alerted with an error message, but return invalid card error to the controller to process with other card business rules on APDU level, and process the alert on controller level.
		offlineApprove[2]	Optional: To allow payment terminal to approve EMV transaction in offline condition below: 0 - Always online. Connection/Timeout error will end with declined.

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			1 - Auto Offline. If connection/Timeout detected, transaction will be stored in batch file for delay online approval. 2- Always Offline. Always approved offline, transactions will be stored in batch files for delay online approval. Delay online authorization is scheduled and automated. NOTE: Enable offlineApprove (Delay authorization) is on MERCHANT RISK. There is no guarantee that delayed authorization will be approved.
signature	64	0E5962DAC59DA96552E00BF7FD0333FA7E49C9011EE465948AD92C35BE80A706	

```
{
  "apiVersion":"1.0",
  "message":"initTerminal",
  "type":"request",
  "timestamp":"2023-05-30 11:06:02",
  "messageTraceID":"112233445566",
  "body":{
    "plazaID":"P01",
    "laneID":"L01",
    "operationMode":"1",
    "laneType":"3"
  },
  "signature":"0E5962DAC59DA96552E00BF7FD0333FA7E49C9011EE465948AD92C35BE80A706"
}
```

Response

Key Name	Max Length	Value	
apiVersion	12	1.0	
message	20	initTerminal	
type	10	ack	
timestamp	20	2023-05-30 11:06:02	
messageTraceID	40	aa11bb22cc33dd44	
body	n	errorCode[4]	0000
signature	64	ADE4FC34CEBCB47B9493ED1277F038E90A7DC7EC4ADB3125D2BA38314ACE60FA	

```
{
  "apiVersion":"1.0",
  "message":"initTerminal",
  "type":"ack",
  "timestamp":"2023-05-30 11:06:02",
  "messageTraceID":"112233445566",
  "body":{
    "errorCode":"0000"
  },
  "signature":"ADE4FC34CEBCB47B9493ED1277F038E90A7DC7EC4ADB3125D2BA38314ACE60FA"
}
```

4.2 Get Status

This function is to get the current reader status and the previous response. Kiosk is required to send this command to get the previous response if the current command is timeout.

Request

Key Name	Max Length	Value
apiVersion	12	1.0
message	20	getStatus
type	10	request
timestamp	20	2023-05-30 11:06:02
messageTraceID	40	112233445566
body	n	empty
signature	64	48E412867FA461765876EF0903233E857823D28BB52169E7DE1EC1C3CB3CFBCB

```
{
  "apiVersion":"1.0",
  "message":"getStatus",
  "type":"request",
  "timestamp":"2023-05-30 11:06:02",
  "messageTraceID":"112233445566",
  "body":{},
  "signature":"48E412867FA461765876EF0903233E857823D28BB52169E7DE1EC1C3CB3CFBCB"
}
```

Response

Key Name	Max Length	Value	
apiVersion	12	1.0	
message	20	getStatus	
type	10	ack	
timestamp	20	2023-05-30 11:06:02	
messageTraceID	40	112233445566	
body	n	errorCode	0000
		operationMode	0 - Maintenance Mode 1 - Live Mode 2 - Not In Use
		state	01 - Idle 02 - Scanning Card 03 - Card Detecting 04 - Card Detected 05 - Host Authorizing 03 - Transaction Completed 99 - Not In Use
		lastCmd	Full command from previous instruction
		lastResp	Full response from previous instruction
signature	64	BE6C458580F8C6256E72D4FBE62C5F6A42BD37470D82273374285B541DDC124F	

```

{
  "apiVersion": "1.0",
  "message": "getStatus",
  "type": "ack",
  "timestamp": "2023-05-30 11:11:21",
  "messageTraceID": "112233445588",
  "body": {
    "errorCode": "0000",
    "operationMode": "1",
    "state": "1",
    "lastCmd": {
      "apiVersion": "1.0",
      "message": "initTerminal",
      "type": "request",
      "timestamp": "2023-05-30 11:06:02",
      "messageTraceID": "112233445566",
      "body": {
        "plazaID": "P01", "laneID": "L01", "operationMode": "1", "laneType": "3"
      }
    },
    "signature": "0E5962DAC59DA96552E00BF7FD0333FA7E49C9011EE465948AD92C35BE80A706"
  },
  "lastResp": {
    "apiVersion": "1.0",
    "message": "initTerminal",
    "type": "request",
    "timestamp": "2023-05-30 11:06:02",
    "messageTraceID": "112233445566",
    "body": {

```

```
        "errorCode": "0000"  
      },  
      "signature": "ADE4FC34CEBCB47B9493ED1277F038E90A7DC7EC4ADB3125D2BA38314ACE60FA"  
    }  
  },  
  "signature": "BE6C458580F8C6256E72D4FBE62C5F6A42BD37470D82273374285B541DDC124F"  
}
```


4.3 Init Card

This function is to start the process of detecting and reading a card. If the card is found valid, it will send an acknowledgement to the kiosk. Kiosk is required to send this command on every end of the transaction if retrigger set to 1, otherwise the reader will go back to detecting and reading a card after Finish Txn. This function is optional.

Request

Key Name	Max Length	Value	
apiVersion	12	1.0	
message	20	initCard	
type	10	request	
timestamp	20	2023-05-30 11:06:02	
messageTraceID	40	112233445566	
body	n	fareClass[2]	This field to be defined by kiosk
		retrigger[1]	0 - Do not required to send initCard. Reader will start scanning card after finTxn. 1 - Required to send initCard to start scanning card.
		title[18]	Optional. Title text to display
		message[40]	Optional. Message text to display
signature	64	0E5962DAC59DA96552E00BF7FD0333FA7E49C9011EE465948AD92C35BE80A706	

```
{
  "apiVersion": "1.0",
  "message": "initCard",
  "type": "request",
  "timestamp": "2023-05-30 11:06:02",
  "messageTraceID": "112233445566",
  "body": {
    "fareClass": "1",
    "retrigger": "1",
  },
  "signature": "0E5962DAC59DA96552E00BF7FD0333FA7E49C9011EE465948AD92C35BE80A706"
}
```

Response

Key Name	Max Length	Value	
apiVersion	12	1.0	
message	20	initCard	
type	10	ack	
timestamp	20	2023-05-30 11:06:02	
messageTraceID	40	aa11bb22cc33dd44	
body	n	errorCode[4]	0000
signature	64	ADE4FC34CEBCB47B9493ED1277F038E90A7DC7EC4ADB3125D2BA38314ACE60FA	

```
{
  "apiVersion":"1.0",
  "message":"initCard",
  "type":"ack",
  "timestamp":"2023-05-30 11:06:02",
  "messageTraceID":"112233445566",
  "body":{
    "errorCode":"0000"
  },
  "signature":"ADE4FC34CEBCB47B9493ED1277F038E90A7DC7EC4ADB3125D2BA38314ACE60FA"
}
```

4.4 Init Entry

This function is to start the process of detecting and reading a card for entry. If the card is found valid, it will send an Init Entry Status to the kiosk. Kiosk is required to send the Proceed Entry command to complete the entry process.

Request

Key Name	Max Length	Value	
apiVersion	12	1.0	
message	20	initEntry	
type	10	request	
timestamp	20	2023-05-30 11:06:02	
messageTraceID	40	112233445566	
body	n	mode[1]	0 - Pre-authorization 1 - Sale Reversal 2 - Card Validate
		fareAmount[12]	350(equal to RM3.50)
		fareClass[2]	This field to be defined by kiosk
		title[18]	Optional. Title text to display
		message[40]	Optional. Message text to display
signature	64	0E5962DAC59DA96552E00BF7FD0333FA7E49C9011EE465948AD92C35BE80A706	

```
{
  "apiVersion":"1.0",
  "message":"initEntry",
  "type":"request",
  "timestamp":"2023-05-30 11:06:02",
  "messageTraceID":"112233445566",
  "body":{
    "mode":"1",
    "fareAmount":"350",
    "fareClass":"1"
  },
  "signature":"0E5962DAC59DA96552E00BF7FD0333FA7E49C9011EE465948AD92C35BE80A706"
}
```

Response

Key Name	Max Length	Value	
apiVersion	12	1.0	
message	20	initEntry	
type	10	ack	
timestamp	20	2023-05-30 11:06:02	
messageTraceID	40	aa11bb22cc33dd44	
body	n	errorCode[4]	0000
signature	64	ADE4FC34CEBCB47B9493ED1277F038E90A7DC7EC4ADB3125D2BA38314ACE60FA	

```
{
  "apiVersion":"1.0",
  "message":"initEntry",
  "type":"ack",
  "timestamp":"2023-05-30 11:06:02",
  "messageTraceID":"aa11bb22cc33dd44",
  "body":{
    "errorCode":"0000"
  },
  "signature":"ADE4FC34CEBCB47B9493ED1277F038E90A7DC7EC4ADB3125D2BA38314ACE60FA"
}
```

4.5 Init Entry Status

This function is sent from reader to kiosk when card is detected and the status of the card (i.e valid card, blacklisted, entry record found/not found).

Request

Key Name	Max Length	Value	
apiVersion	12	1.0	
message	20	initEntryStatus	
type	10	request	
timestamp	20	2023-05-30 11:06:02	
messageTraceID	40	aa11bb22cc33dd44	
body	n	errorCode[4]	0000
		entryDt[20]	2023-05-30 11:06:02
		maskPan[20]	Masked PAN. Only first 6 and last 4 will be returned in clear, others will be masked with *. 4935 31** **** 2683 Non EMV will be returned in clear. 1000140007140828
		hashPan[40]	Hashed PAN. Hash with a full clear Card PAN. 649B28A9F095B9716AA9543067ED4F9A995E8F62
		cardScheme[2]	01 - VISA 02 - MasterCard 03 - MyDebit
		status[2]	Entry Status 01 - Entry found in system 02 - Entry not found in system
		cardBal[12]	Card Balance. TNG Card Only 39979 - \$399.79
		mfgNo[]	Mfg No. TNG Card Only 0286144655
		cardTxnNo[]	Card Txn No. TNG Card Only 62979

[illegible]

```
{
  "apiVersion": "1.0",
  "message": "initEntryStatus",
  "type": "request",
  "timestamp": "2023-05-30 11:06:02",
  "messageTraceID": "112233445577",
  "body": {
    "errorCode": "0000",
    "entryDt": "2023-05-30 11:06:02",
    "maskPan": "493531*****2683",
    "hashPan": "649B28A9F095B9716AA9543067ED4F9A995E8F62",
    "cardScheme": "01"
    "Status": "01",
  },
  "signature": "5FFF82589F9FF7C6A861FED738A069628EBB8F24DDE56A9AFA8BEFB990CFED97"
}
```

}

Response

Key Name	Max Length	Value	
apiVersion	12	1.0	
message	10	initEntryStatus	
type	10	ack	
timestamp	20	2023-05-30 11:06:02	
messageTraceID	40	112233445577	
body	n	errorCode	0000
signature	64		

```
{
  "apiVersion":"1.0",
  "message":"initEntryStatus",
  "type":"ack",
  "timestamp":"2023-05-30 11:06:02",
  "messageTraceID":"112233445577",
  "body":{
    "errorCode":"0000"
  },
  "signature":"5FFF82589F9FF7C6A861FED738A069628EBB8F24DDE56A9AFA8BEFB990CFED97"
}
```


4.6 Proceed Entry

This function is to authorize the transaction with the host to complete the entry transaction. Transaction response will be returned to the kiosk.

Request

Key Name	Max Length	Value	
apiVersion	12	1.0	
message	20	proceedEntry	
type	10	request	
timestamp	20	2023-05-30 11:06:02	
messageTraceID	40	112233445577	
body	n	payFlag	Payment Flag. This field is optional. Eg: 0 - Tng Purse 1 - Tng E-Wallet
signature	64	5FFF82589F9FF7C6A861FED738A069628EBB8F24DDE56A9AFA8BEFB990CFED97	

```
{
  "apiVersion": "1.0",
  "message": "proceedEntry",
  "type": "request",
  "timestamp": "2023-05-30 11:06:02",
  "messageTraceID": "112233445577",
  "body": {},
  "signature": "5FFF82589F9FF7C6A861FED738A069628EBB8F24DDE56A9AFA8BEFB990CFED97"
}
```

Response

Key Name	Max Length	Value	
apiVersion	12	1.0	
message	10	proceedEntry	
type	10	ack	
timestamp	20	2023-05-30 11:06:02	
messageTraceID	40	112233445577	
body	n	errorCode	0000
		txnID	Terminal transaction ID. Generated from terminal. 0A6CDF1690177983701
		paymentMode	Transaction Payment Mode. online/offline
		entryDt[20]	2023-05-30 11:06:02
		maskPan[20]	Masked PAN. EMV only first 6 and last 4 will be returned in clear, others will be masked with *. 4935 31** ***** 2683 Non EMV will be returned in clear. 1000140007140828
		hashPan[40]	Hashed PAN. Hash with a full clear Card PAN. 649B28A9F095B9716AA9543067ED4F9A995E8F62
		cardScheme[2]	01 - VISA 02 - MasterCard 03 - MyDebit 11 - TNG
		stan[6]	System Trace Number. Sent from reader to acquiring host. 000001
		rrn[12]	Retrieval Reference Number. Returned from acquiring host. 000012300001
		apprCode[6]	Approval Code. Returned from acquiring host. 123456

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[illegible]

		Decode: Sector - 00 Block - 01 Value - 000102030405060708090A0B0C0D0E0F
signature	64	

```
{
  "apiVersion": "1.0",
  "message": "proceedEntry",
  "type": "ack",
  "timestamp": "2023-05-30 11:06:02",
  "messageTraceID": "112233445577",
  "body": {
    "errorCode": "0000",
    "txnId": "0A6CDF1690177983701",
    "paymentMode": "online",
    "entryDt": "2023-05-30 11:06:02",
    "maskPan": "493531*****2683",
    "hashPan": "649B28A9F095B9716AA9543067ED4F9A995E8F62",
    "cardScheme": "01",
    "stan": "000001",
    "rrn": "000012300001",
    "apprCode": "123456",
    "invNo": "000001",
    "fareAmount": "100",
    "fareClass": "1",
  },
  "signature": "5FFF82589F9FF7C6A861FED738A069628EBB8F24DDE56A9AFA8BEFB990CFED97"
}
```

[illegible]

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```
"signature":"01F70F9148BA8D3474CCB764CA0233230902719C06ADF5238ECA475465CF"
```

4.7 Init Exit

This function is to start the process of detecting and reading a card for exit. Mode must be the same as the entry mode. If the card is found valid and entry record is found, it will send an Init Exit Status to the kiosk. Kiosk is required to send the Proceed Exit command with fare to complete the exit process.

Request

Key Name	Max Length	Value	
apiVersion	12	1.0	
message	20	initExit	
type	10	request	
timestamp	20	2023-05-30 11:06:02	
messageTraceID	40	112233445566	
body	n	mode[1]	0 - Pre-authorization 1 - Sale Reversal 2 - Card Validate
		title[18]	Optional. Title text to display
		message[40]	Optional. Message text to display
signature	64	0E5962DAC59DA96552E00BF7FD0333FA7E49C9011EE465948AD92C35BE80A706	

```
{
  "apiVersion":"1.0",
  "message":"initExit",
  "type":"request",
  "timestamp":"2023-05-30 11:06:02",
  "messageTraceID":"112233445566",
  "body":{
    "mode":"1"
  },
  "signature":"0E5962DAC59DA96552E00BF7FD0333FA7E49C9011EE465948AD92C35BE80A706"
}
```

Response

Key Name	Max Length	Value	
apiVersion	12	1.0	
message	20	initExit	
type	10	ack	
timestamp	20	2023-05-30 11:06:02	
messageTraceID	40	aa11bb22cc33dd44	
body	n	errorCode[4]	0000
signature	64	ADE4FC34CEBCB47B9493ED1277F038E90A7DC7EC4ADB3125D2BA38314ACE60FA	

```
{
  "apiVersion":"1.0",
  "message":"initEntry",
  "type":"ack",
  "timestamp":"2023-05-30 11:06:02",
  "messageTraceID":"112233445566",
  "body":{
    "errorCode":"0000"
  },
  "signature":"ADE4FC34CEBCB47B9493ED1277F038E90A7DC7EC4ADB3125D2BA38314ACE60FA"
}
```

4.8 Init Exit Status

This function is sent from reader to kiosk when card is detected and the status of the card (i.e valid card, blacklisted, entry record found/not found).

Request

Key Name	Max Length	Value	
apiVersion	12	1.0	
message	20	initExitStatus	
type	10	request	
timestamp	20	2023-05-30 11:06:02	
messageTraceID	40	aa11bb22cc33dd44	
body	n	errorCode[4]	0000
		entryDt[20]	2023-05-30 11:06:02
		exitDt[20]	2023-05-30 11:06:02
		maskPan[20]	Masked PAN. Only first 6 and last 4 will be returned in clear, others will be masked with *. 4935 31** **** 2683 Non EMV will be returned in clear. 1000140007140828
		hashPan[40]	Hashed PAN. Hash with a full clear Card PAN. 649B28A9F095B9716AA9543067ED4F9A995E8F62
		cardScheme[2]	01 - VISA 02 - MasterCard 03 - MyDebit
		status[2]	Entry Status 01 - Entry found in system 02 - Entry not found in system
		cardBal[12]	Card Balance. TNG Card Only 39979 - \$399.79
		mfgNo[]	Mfg No. TNG Card Only 0286144655
		cardTxnNo[]	Card Txn No. TNG Card Only

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		62979
	purseFlag[]	Purse Flag. TNG Card Only P - Purse E - E-wallet
	machineCode	Machine Code. TNG Card Only 41424344
	cardNo	Card No. TNG Card Only 601464000788007155
	corporateFlag	Corporate Flag. TNG Card Only N
	userCat	User Cat. TNG Card Only N
	cscType	Csc Type. TNG Card Only R
	chipData	Parking Chip Data. TNG Card Only 0001000102030405060708090A0B0C0D0E0F000200C44EBA0 00402177E400004B2674B050500000000000000000000000000000000 000000000050100 000 000624044AF0001A00702A082E044AF00010200000000000000000000 00000D00414243445000330000000000000000000000000000000000 3B69C5BBAEDF04902003530F99C0E000000000000000000000000000 0000000000000000 Decode: Sector - 00 Block - 01 Value - 000102030405060708090A0B0C0D0E0F
signature	64	5FFF82589F9FF7C6A861FED738A069628EBB8F24DDE56A9AFA8BEFB990CFED97

EMV

```
{
  "apiVersion": "1.0",
  "message": "initExitStatus",
  "type": "request",
  "timestamp": "2023-05-30 11:06:02",
  "messageTraceID": "112233445577",
  "body": {
    "errorCode": "0000",

    "entryDt": "2023-05-30 11:06:02",
    "exitDt": "2023-05-30 11:06:02",
    "maskPan": "493531*****2683",
    "hashPan": "649B28A9F095B9716AA9543067ED4F9A995E8F62",
    "cardScheme": "01"
    "status": "01",
  }
}
```


Response

Key Name	Max Length	Value	
apiVersion	12	1.0	
message	10	initExitStatus	
type	10	ack	
timestamp	20	2023-05-30 11:06:02	
messageTraceID	40	112233445577	
body	n	errorCode	0000
signature	64		

```
{
  "apiVersion": "1.0",
  "message": "initExitStatus",
  "type": "ack",
  "timestamp": "2023-05-30 11:06:02",
  "messageTraceID": "112233445577",
  "body": {
    "errorCode": "0000"
  },
  "signature": "5FFF82589F9FF7C6A861FED738A069628EBB8F24DDE56A9AFA8BEFB990CFED97"
}
```

4.9 Proceed Exit

This function is to authorize the transaction with the host to complete the exit transaction. Transaction response will be returned to the kiosk.

Request

Key Name	Max Length	Value	
apiVersion	12	1.0	
message	20	proceedExit	
type	10	request	
timestamp	20	2023-05-30 11:06:02	
messageTraceID	40	112233445577	
body	n	fareAmount[12]	350 (equal to RM 3.50)
		fareClass[2]	This field to be defined by kiosk
		fallTimeout[3]	Fallback Timeout. This will start counting when exit transaction is failed. Transaction status will be send in Proceed Exit Status. 0 - Disable fallback 60 - Accept fallback card within 60 seconds
		entryDt[20]	Entry Date Time. 2024-07-12 16:24
		payFlag	Payment Flag. This field is optional. Eg: 0 - Tng Purse 1 - Tng E-Wallet
signature	64	5FFF82589F9FF7C6A861FED738A069628EBB8F24DDE56A9AFA8BEFB990CFED97	

```
{
  "apiVersion": "1.0",
  "message": "proceedExit",
  "type": "request",
  "timestamp": "2023-05-30 11:06:02",
  "messageTraceID": "112233445577",
  "body": {
    "fareAmount": "100",
    "fareClass": "1",
    "fallTimeout": "0",
    "entryDt": "2024-07-12 16:24"
  },
  "signature": "5FFF82589F9FF7C6A861FED738A069628EBB8F24DDE56A9AFA8BEFB990CFED97"
}
```

Response

Key Name	Max Length	Value	
apiVersion	12	1.0	
message	20	initExit	
type	10	ack	
timestamp	20	2023-05-30 11:06:02	
messageTraceID	40	aa11bb22cc33dd44	
body	n	errorCode[4]	0000
signature	64	ADE4FC34CEBCB47B9493ED1277F038E90A7DC7EC4ADB3125D2BA38314ACE60FA	

```
{
  "apiVersion":"1.0",
  "message":"proceedExit",
  "type":"ack",
  "timestamp":"2023-05-30 11:06:02",
  "messageTraceID":"112233445566",
  "body":{
    "errorCode":"0000"
  },
  "signature":"ADE4FC34CEBCB47B9493ED1277F038E90A7DC7EC4ADB3125D2BA38314ACE60FA"
}
```

4.10 Proceed Exit Status

This function is sent from reader to kiosk on the status of the transaction with the host to complete the exit transaction.

Request

Key Name	Max Length	Value	
apiVersion	12	1.0	
message	20	proceedExitStatus	
type	10	request	
timestamp	20	2023-05-30 11:06:02	
messageTraceID	40	aa11bb22cc33dd44	
body	n	errorCode[4]	0000
		entryDt[20]	2023-05-30 11:06:02
		exitDt[20]	2023-05-30 11:06:02
		maskPan[20]	Masked PAN. Only first 6 and last 4 will be returned in clear, others will be masked with *. 4935 31** **** 2683 Non EMV will be returned in clear. 1000140007140828
		hashPan[40]	Hashed PAN. Hash with a full clear Card PAN. 649B28A9F095B9716AA9543067ED4F9A995E8F62
		cardScheme[2]	01 - VISA 02 - MasterCard 03 - MyDebit
		status[2]	Entry Status 01 - Entry found in system 02 - Entry not found in system
		cardBal[12]	Card Balance. TNG Card Only 39979 - \$399.79
		mfgNo[]	Mfg No. TNG Card Only 0286144655
		cardTxnNo[]	Card Txn No. TNG Card Only

Response

Key Name	Max Length	Value	
apiVersion	12	1.0	
message	10	proceedExitStatus	
type	10	ack	
timestamp	20	2023-05-30 11:06:02	
messageTraceID	40	112233445577	
body	n	errorCode	0000
signature	64		

```
{
  "apiVersion":"1.0",
  "message":"proceedExitStatus",
  "type":"ack",

  "timestamp":"2023-05-30 11:06:02",
  "messageTraceID":"112233445577",
  "body":{
    "errorCode":"0000"
  },
  "signature":"5FFF82589F9FF7C6A861FED738A069628EBB8F24DDE56A9AFA8BEFB990CFED97"
}
```

4.11 Start Transaction

This function read card, validates internally (i.e blacklist, whitelist, expiry check). If the card is found valid, it will authorize acquiring or deducting the card purse balance. TxnStatus will be sent from the reader to kiosk on the status of the transaction.

Request

Key Name	Max Length	Value	
apiVersion	12	1.0	
message	20	startTxn	
type	10	request	
timestamp	20	2023-05-30 11:06:02	
messageTraceID	40	112233445566	
body	n	txnType[12]	Transaction type of this transaction. 1. Entry 2. Exit 3. Dual 4. Single
		fareAmount[12]	350 (equal to RM 3.50)
		fareClass[2]	This field to be defined by kiosk
		title[18]	Optional. Title text to display
		message[40]	Optional. Message text to display
signature	64	1BE1581D1CA63EBD65D919DFFB7F575DDC396A5C4822CE6F395C1BEC34010E39	

```
{
  "apiVersion":"1.0",
  "message":"startTxn",
  "type":"request",
  "timestamp":"2023-05-30 11:06:02",
  "messageTraceID":"112233445566",
  "Body":{
    "txnType":"single",
    "fareAmount":"350",
    "fareClass":"1",
  },
  "signature":"1BE1581D1CA63EBD65D919DFFB7F575DDC396A5C4822CE6F395C1BEC34010E39"
}
```

Response

Key Name	Max Length	Value	
apiVersion	12	1.0	
message	10	startTxn	
type	10	ack	
timestamp	20	2023-05-30 11:07:32	
messageTraceID	40	112233445566	
body	n	errorCode	0000
signature	64	29DD4219279A020368A102BBF57A2EBA958BC43D8EC7A4F8EB04E9C063A6D075	

```
{
  "apiVersion":"1.0",
  "message":"startTxn",
  "type":"ack",
  "timestamp":"2023-05-30 11:07:32",
  "messageTraceID":"112233445566",
  "body":{
    "errorCode":"0000"
  },
  "signature":"29DD4219279A020368A102BBF57A2EBA958BC43D8EC7A4F8EB04E9C063A6D075"
}
```

4.12 Abort Txn

This function is to abort the card scanning process and terminate the transaction and return to idle state.

Request

Key Name	Max Length	Value	
apiVersion	12	1.0	
message	20	abortTxn	
type	10	request	
timestamp	20	2023-05-30 11:06:02	
messageTraceID	40	112233445566	
body	n	title[18]	Optional. Title text to display
		message[40]	Optional. Message text to display
signature	64	4083E507079483AB35B37794D366E0EFE6BFC1647E10FA9FCA1CBE3C8E200DD1	

```
{
  "apiVersion":"1.0",
  "message":"abortTxn",
  "type":"request",
  "timestamp":"2023-05-30 11:06:02",
  "messageTraceID":"112233445566",
  "body":{ },
  "signature":"4083E507079483AB35B37794D366E0EFE6BFC1647E10FA9FCA1CBE3C8E200DD1"
}
```

Response

Key Name	Max Length	Value	
apiVersion	12	1.0	
message	10	abortTxn	
type	10	ack	
timestamp	20	2023-05-30 11:06:02	
messageTraceID	40	112233445566	
body	n	errorCode	0000
signature	64	79C591841928BC2010E247729C6873C30206BA2B50F26261C4588E300F71DF02	

```
{
  "apiVersion":"1.0",
  "message":"abortTxn",
  "type":"ack",
  "timestamp":"2023-05-30 11:22:54",
  "messageTraceID":"112233445566",
  "body":{
    "errorCode":"0000"
  },
  "signature":"79C591841928BC2010E247729C6873C30206BA2B50F26261C4588E300F71DF02"
}
```

4.13 Txn Status

This function is sent from the reader to kiosk on the status of the transaction for Start Txn (i.e approved, declined from the acquiring host).

Request

Key Name	Max Length	Value	
apiVersion	12	1.0	
message	20	txnStatus	
type	10	request	
timestamp	20	2023-05-30 11:06:02	
messageTraceID	40	112233445577	
body	n	errorCode[4]	0000
		txnID	Terminal transaction ID. Generated from terminal. 0A6CDF1690177983701
		txnDt[20]	2023-05-30 11:06:02
		maskPan[20]	Masked PAN. EMV only first 6 and last 4 will be returned in clear, others will be masked with *. 4935 31** **** 2683 Non EMV will be returned in clear. 1000140007140828
		hashPan[40]	Hashed PAN. Hash with a full clear Card PAN. 649B28A9F095B9716AA9543067ED4F9A995E8F62
		cardScheme[2]	01 - VISA 02 - MasterCard 03 - MyDebit 11 - Tng 12 - Ezlink 13 - Nets
		stan[6]	System Trace Number. Sent from reader to acquiring host. 000001
		rrn[12]	Retrieval Reference Number. Returned from acquiring host. 000012300001

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		apprCode[6]	Approval Code. Returned from acquiring host. 123456
		invNo[6]	Invoice Number. Sent from reader to acquiring host. 000001
		fareAmount	Fare Amount. 100 (equal to RM1.00)
		fareClass	Fare Class. 1
		cardBal	Card Balance. Only for non emv card. 10321
signature	64	5FFF82589F9FF7C6A861FED738A069628EBB8F24DDE56A9AFA8BEFB990CFED97	

```
{
  "apiVersion":"1.0",
  "message":"txnStatus",
  "type":"request",
  "timestamp":"2023-05-30 11:06:02",
  "messageTraceID":"112233445577",
  "body":{
    "errorCode":"0000",
    "txnID":"0A6CDF1690177983701",
    "txnDt":"2023-05-30 11:06:02",
    "maskPan":"493531*****2683",
    "hashPan":"649B28A9F095B9716AA9543067ED4F9A995E8F62",
    "cardScheme":"01",
    "stan":"000001",
    "rrn":"000012300001",
    "apprCode":"123456",
    "invNo":"000001",
    "fareAmount":"100",
    "fareClass":"1",
    "cardBal":"10321"
  },
  "signature":"5FFF82589F9FF7C6A861FED738A069628EBB8F24DDE56A9AFA8BEFB990CFED97"
}
```

Response

Key Name	Max Length	Value	
apiVersion	12	1.0	
message	10	txnStatus	
type	10	ack	
timestamp	20	2023-05-30 11:06:02	
messageTraceID	40	112233445577	
body	n	errorCode	0000
signature	64		

```
{
  "apiVersion":"1.0",
  "message":"txnStatus",
  "type":"ack",
  "timestamp":"2023-05-30 11:06:02",
  "messageTraceID":"112233445577",
  "body":{
    "errorCode":"0000"
  },
  "signature":"5FFF82589F9FF7C6A861FED738A069628EBB8F24DDE56A9AFA8BEFB990CFED97"
}
```


4.14 Txn Query

This function is to query the offline transaction posting status. Only valid offline transactions in the batch will be returned.

Request

Key Name	Max Length	Value	
apiVersion	12	1.0	
message	20	txnQuery	
type	10	request	
timestamp	20	2023-05-30 11:06:02	
messageTraceID	40	112233445566	
body	n	txnID[18]	Terminal transaction ID. Return from txnStatus.
signature	64	4083E507079483AB35B37794D366E0EFE6BFC1647E10FA9FCA1CBE3C8E200DD1	

```
{
  "apiVersion":"1.0",
  "message":"txnQuery",
  "type":"request",
  "timestamp":"2023-05-30 11:06:02",
  "messageTraceID":"112233445566",
  "body":{
    "txnID":"0A6CDF1690177983701"
  },
  "signature":"4083E507079483AB35B37794D366E0EFE6BFC1647E10FA9FCA1CBE3C8E200DD1"
}
```

Response

Key Name	Max Length	Value	
apiVersion	12	1.0	
message	10	txnQuery	
type	10	ack	
timestamp	20	2023-05-30 11:06:02	
messageTraceID	40	112233445566	
body	n	errorCode	0000
		txnID	Terminal transaction ID. Send from kiosk
		txnStatus	Transaction Status Transaction approved - approved Transaction rejected - rejected Transaction pending upload - pending
signature	64	79C591841928BC2010E247729C6873C30206BA2B50F26261C4588E300F71DF02	

```
{
  "apiVersion": "1.0",
  "message": "txnQuery",
  "type": "ack",
  "timestamp": "2023-05-30 11:22:54",
  "messageTraceID": "112233445566",
  "body": {
    "errorCode": "0000",
    "txnID": "173349912994442012",
    "txnStatus": "approved"
  },
  "signature": "79C591841928BC2010E247729C6873C30206BA2B50F26261C4588E300F71DF02"
}
```

4.15 Finish Transaction

This function is to complete the transaction. Kiosk sends this command after the transaction is completed / barrier is opened and the reader will refresh the screen. If this command is not received by the reader, the reader screen will remain at the Transaction Completed screen for operation to support.

Request

Key Name	Max Length	Value
apiVersion	12	1.0
message	20	finTxn
type	10	request
timestamp	20	2023-05-30 11:06:02
messageTraceID	40	112233445566
body	n	empty
signature	64	9B21B4060695E406A9BAA5D07FA75DE20338F9C6BB310627EEFF7AB37E143442

```
{
  "apiVersion":"1.0",
  "message":"finTxn",
  "type":"request",
  "timestamp":"2023-05-30 11:06:02",
  "messageTraceID":"112233445566",
  "body":{},
  "signature":"9B21B4060695E406A9BAA5D07FA75DE20338F9C6BB310627EEFF7AB37E143442"
}
```

Response

Key Name	Max Length	Value	
apiVersion	12	1.0	
message	10	finTxn	
type	10	ack	
timestamp	20	2023-05-30 11:06:02	
messageTraceID	40	112233445566	
body	n	errorCode	0000
signature	64	228B6582D373135EA7BB53CA073EB4528C05A35DB38F5DB476F0059910EAFEAD	

```
{
  "apiVersion":"1.0",
  "message":"finTxn",
  "type":"ack",
  "timestamp":"2023-05-30 11:06:04",
  "messageTraceID":"112233445566",
  "body":{
    "errorCode":"0000"
  },
  "signature":"228B6582D373135EA7BB53CA073EB4528C05A35DB38F5DB476F0059910EAFEAD"
}
```

4.16 Set Blacklist (TNG)

This function is to set the blacklist flag for TNG card.

Request

Key Name	Max Length	Value
apiVersion	12	1.0
message	20	setBlacklist
type	10	request
timestamp	20	2023-05-30 11:06:02
messageTraceID	40	112233445566
body	n	empty
signature	64	9B21B4060695E406A9BAA5D07FA75DE20338F9C6BB310627EEFF7AB37E143442

```
{
  "apiVersion": "1.0",
  "message": "setBlacklist",
  "type": "request",
  "timestamp": "2023-05-30 11:06:02",
  "messageTraceID": "112233445566",
  "body": {},
  "signature": "9B21B4060695E406A9BAA5D07FA75DE20338F9C6BB310627EEFF7AB37E143442"
}
```

Response

Key Name	Max Length	Value	
apiVersion	12	1.0	
message	10	setBlacklist	
type	10	ack	
timestamp	20	2023-05-30 11:06:02	
messageTraceID	40	112233445566	
body	n	errorCode	0000
signature	64	228B6582D373135EA7BB53CA073EB4528C05A35DB38F5DB476F0059910EAFEAD	

```
{
  "apiVersion":"1.0",
  "message":"setBlacklist",
  "type":"ack",
  "timestamp":"2023-05-30 11:06:04",
  "messageTraceID":"112233445566",
  "body":{
    "errorCode":"0000"
  },
  "signature":"228B6582D373135EA7BB53CA073EB4528C05A35DB38F5DB476F0059910EAFEAD"
}
```

4.17 Set AutoReloadFlag (TNG)

This function is to set the AutoReload flag for TNG card.

Request

Key Name	Max Length	Value	
apiVersion	12	1.0	
message	20	setAutoReloadFlag	
type	10	request	
timestamp	20	2023-05-30 11:06:02	
messageTraceID	40	112233445566	
body	n	flagValue	0 or 1
signature	64	9B21B4060695E406A9BAA5D07FA75DE20338F9C6BB310627EEFF7AB37E143442	

```
{
  "apiVersion": "1.0",
  "message": "setAutoReloadFlag",
  "type": "request",
  "timestamp": "2023-05-30 11:06:02",
  "messageTraceID": "112233445566",
  "Body": {
    "flagValue": "0"
  },
  "signature": "9B21B4060695E406A9BAA5D07FA75DE20338F9C6BB310627EEFF7AB37E143442"
}
```

Response

Key Name	Max Length	Value	
apiVersion	12	1.0	
message	10	setAutoReloadFlag	
type	10	ack	
timestamp	20	2023-05-30 11:06:02	
messageTraceID	40	112233445566	
body	n	errorCode	0000
signature	64	228B6582D373135EA7BB53CA073EB4528C05A35DB38F5DB476F0059910EAFEAD	

```
{
  "apiVersion":"1.0",
  "message":"setAutoReloadFlag",
  "type":"ack",
  "timestamp":"2023-05-30 11:06:04",
  "messageTraceID":"112233445566",
  "body":{
    "errorCode":"0000"
  },
  "signature":"228B6582D373135EA7BB53CA073EB4528C05A35DB38F5DB476F0059910EAFEAD"
}
```


4.18 Set Reader Ip

This function is to configure the reader ip.

Request

Key Name	Max Length	Value	
apiVersion	12	1.0	
message	20	setReaderIp	
type	10	request	
timestamp	20	2023-05-30 11:06:02	
messageTraceID	40	112233445566	
body	n	ip	192.168.1.199
		netmask	255.255.255.0
		gateway	192.168.1.1
		dhcp	0
signature	64	9B21B4060695E406A9BAA5D07FA75DE20338F9C6BB310627EEFF7AB37E143442	

```
{
  "apiVersion":"1.0",
  "message":"setReaderIp",
  "type":"request",
  "timestamp":"2023-06-26 11:30:30",
  "messageTraceID":"112233445566",
  "body":{
    "ip":"192.168.1.199",
    "netmask":"255.255.255.0",
    "gateway":"192.168.1.1",
    "dhcp":"0"
  },
  "signature":"E58129F980A7957A85CC238D58A0A357D9FD6FCA1D7A8D713CBFAB4237708543"
}
```

Response

Key Name	Max Length	Value	
apiVersion	12	1.0	
message	10	setReaderIp	
type	10	ack	
timestamp	20	2023-05-30 11:06:02	
messageTraceID	40	112233445566	
body	n	errorCode	0000
signature	64	228B6582D373135EA7BB53CA073EB4528C05A35DB38F5DB476F0059910EAFEAD	

```
{
  "apiVersion":"1.0",
  "message":"setReaderIp",
  "type":"ack",
  "timestamp":"2023-06-26 11:32:01",
  "messageTraceID":"112233445566",
  "body":{
    "errorCode":"0000"
  },
  "signature":"69A9CB53658F8ECB9F07C3ECED4EA33F7413B0ED0E1E9657317C89A14DA3A9BF"
}
```

4.19 Reboot

This function is to reboot the reader. To reboot the reader, operation mode must be in “Not In Use”.

Request

Key Name	Max Length	Value
apiVersion	12	1.0
message	20	reboot
type	10	request
timestamp	20	2023-05-30 11:06:02
messageTraceID	40	112233445566
body	n	empty
signature	64	9B21B4060695E406A9BAA5D07FA75DE20338F9C6BB310627EEFF7AB37E143442

```
{
  "apiVersion":"1.0",
  "message":"reboot",
  "type":"request",
  "timestamp":"2023-06-26 11:30:30",
  "messageTraceID":"112233445566",
  "body":{},
  "signature":"5EE94E3BD613AAD3AA9355DAC44704F4654841F453BBF2F00BD969D27E54FE95"
}
```

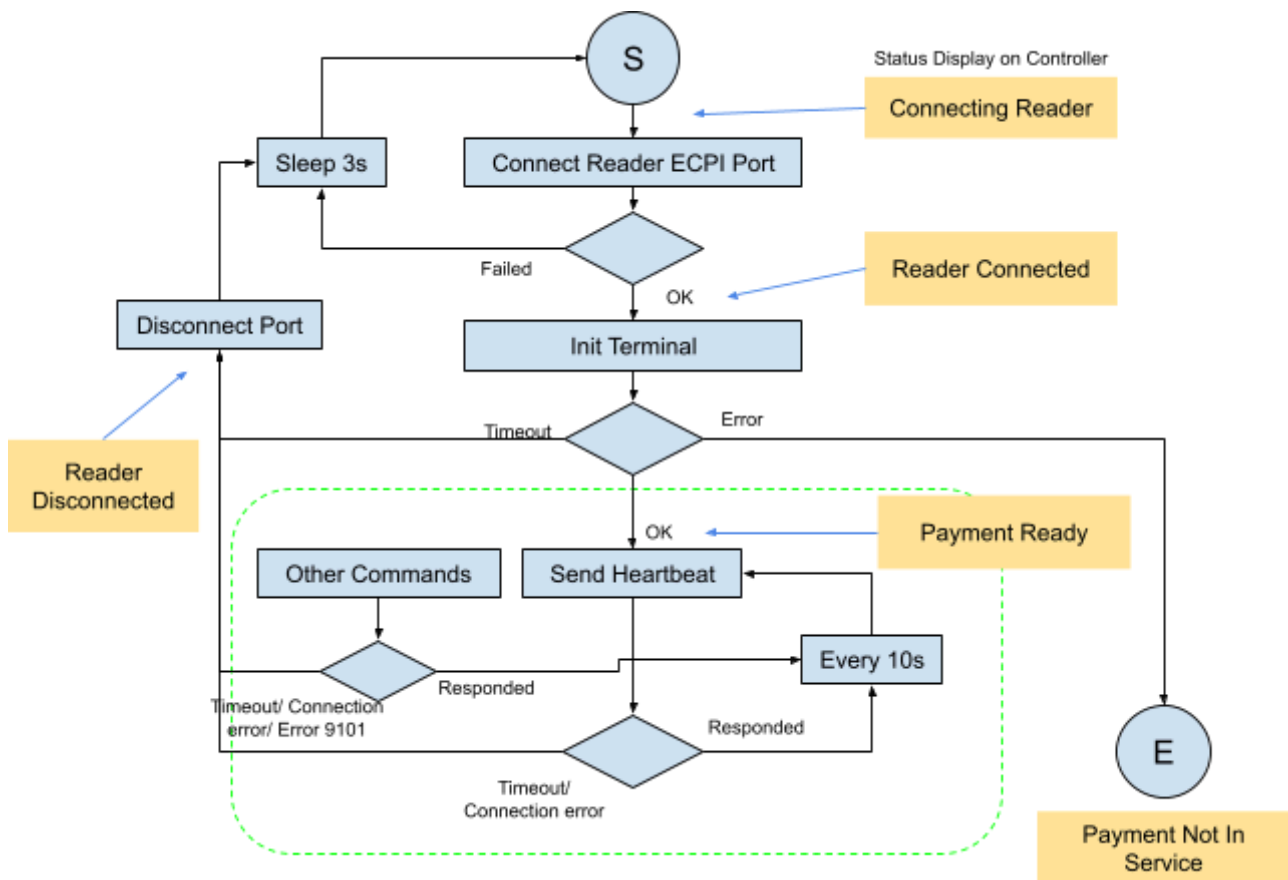
Response

Key Name	Max Length	Value	
apiVersion	12	1.0	
message	10	reboot	
type	10	ack	
timestamp	20	2023-05-30 11:06:02	
messageTraceID	40	112233445566	
body	n	errorCode	0000
signature	64	228B6582D373135EA7BB53CA073EB4528C05A35DB38F5DB476F0059910EAFEAD	

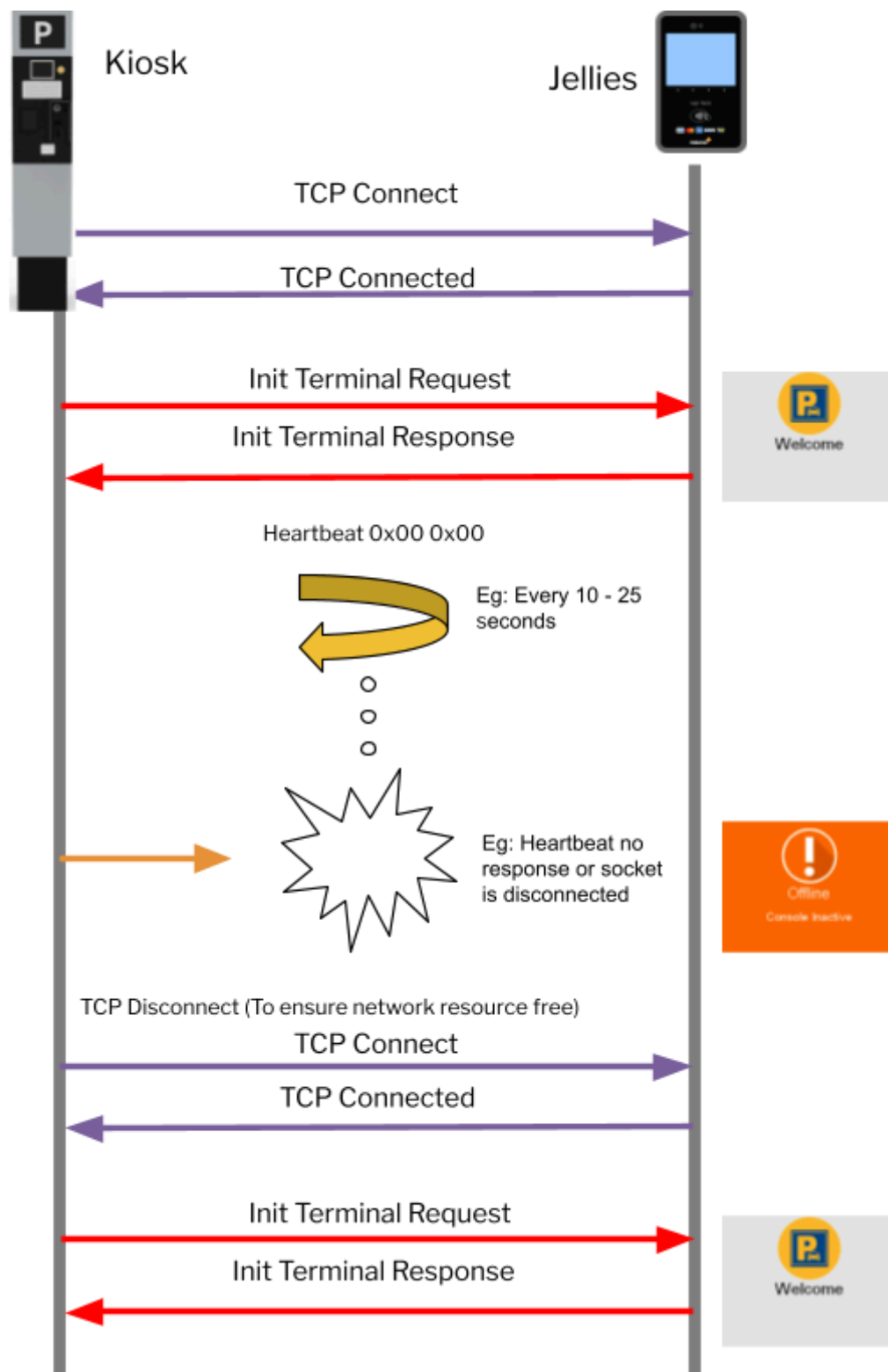
```
{
  "apiVersion":"1.0",
  "message":"reboot",
  "type":"ack",
  "timestamp":"2023-06-26 11:30:32",
  "messageTraceID":"112233445566",
  "body":{
    "errorCode":"0000"
  },
  "signature":"B1491B7B2D1557EB03AE8F85A147163E0A033E16EA7AAC7D2C54DFC49B1041ED"
}
```

5 Process Flow

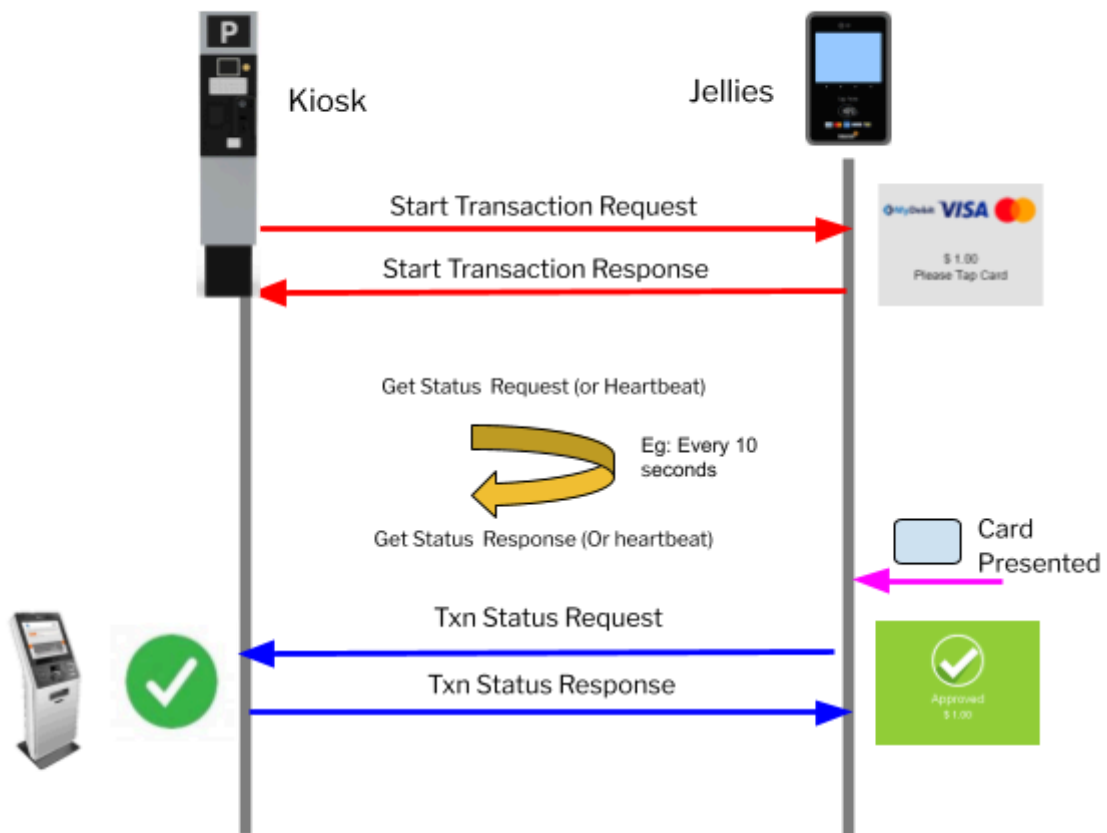
General process flow of controller is demonstrated below:



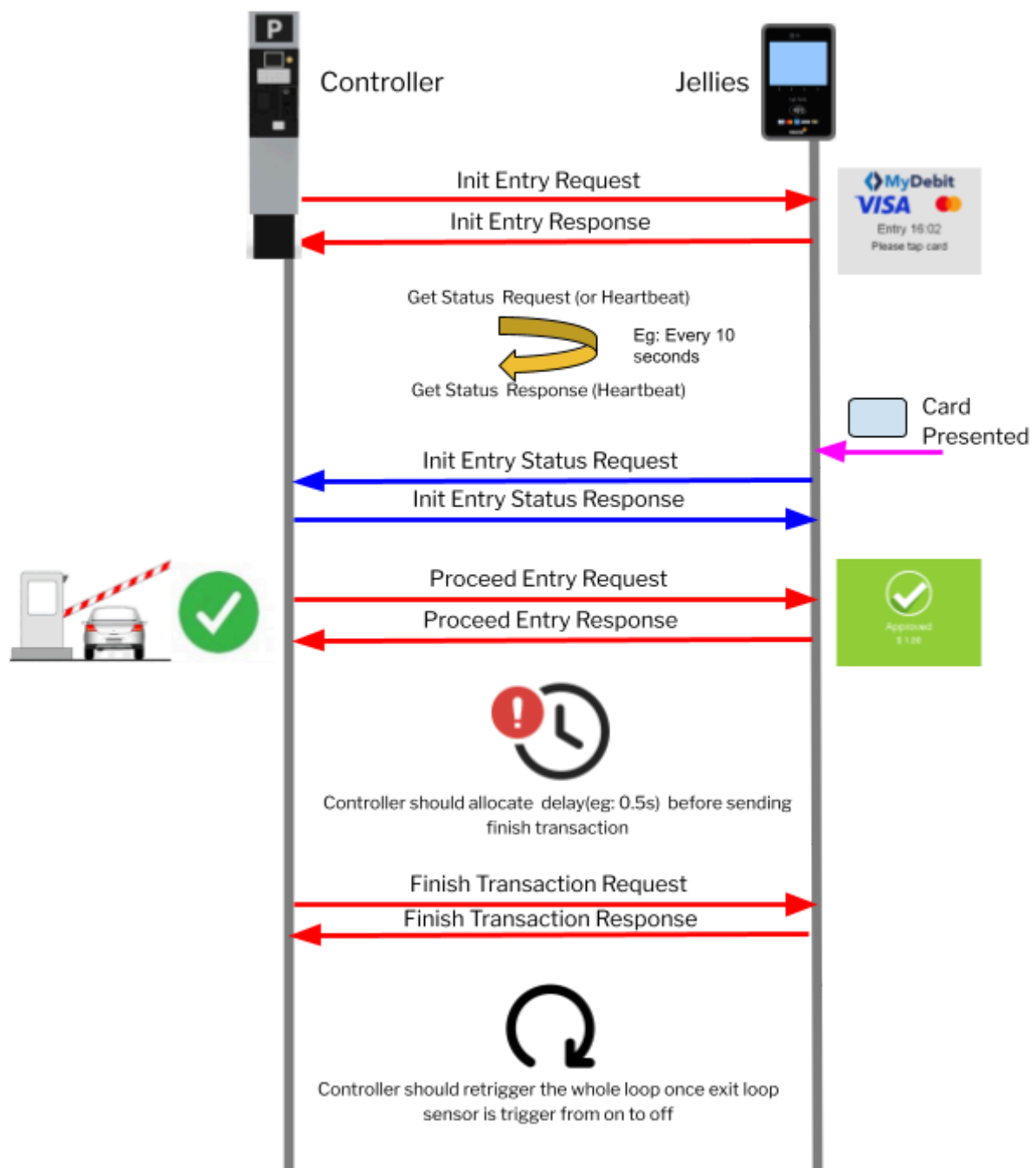
5.1 Initialization



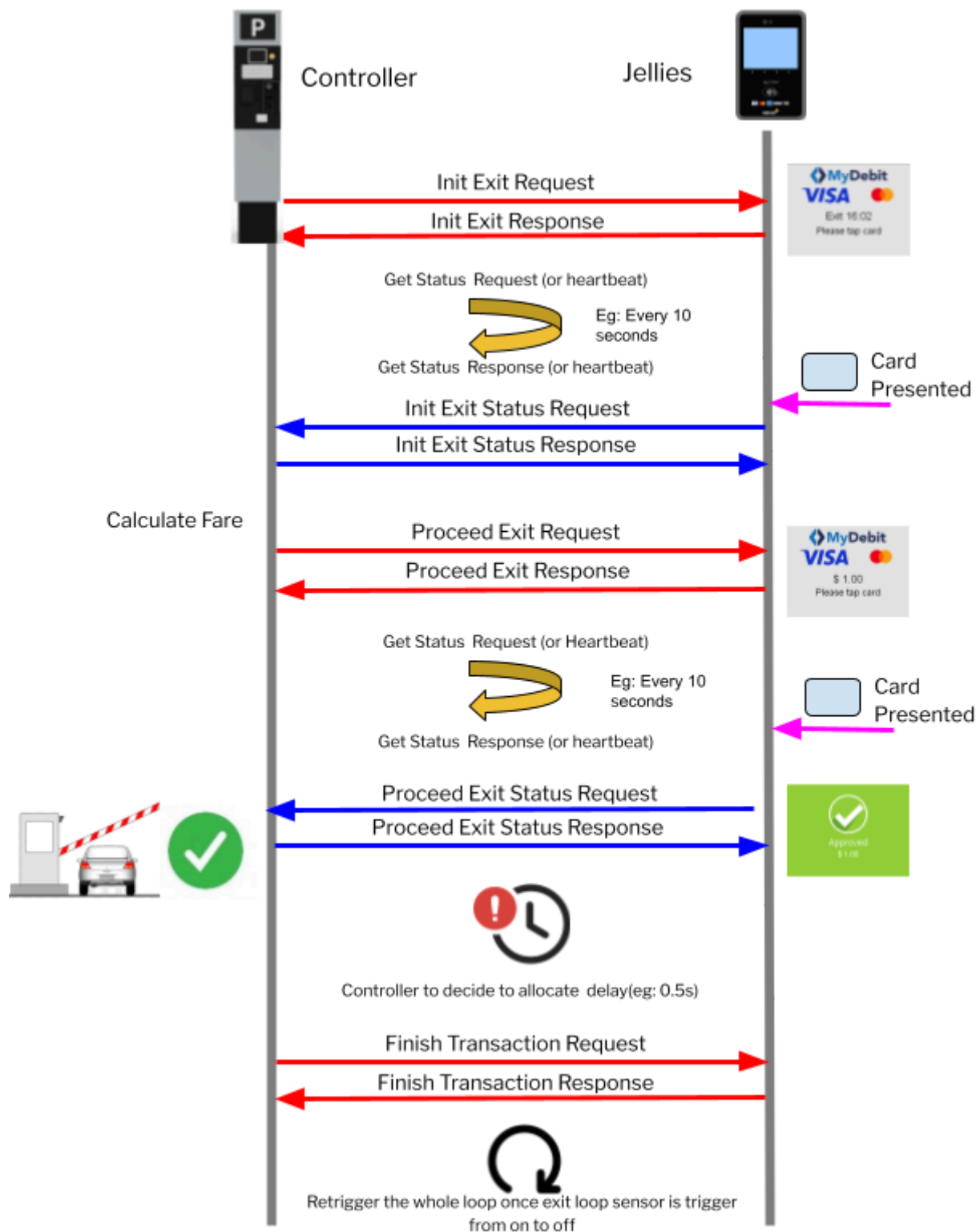
5.2 Sale Transaction



5.3 Parking Entry

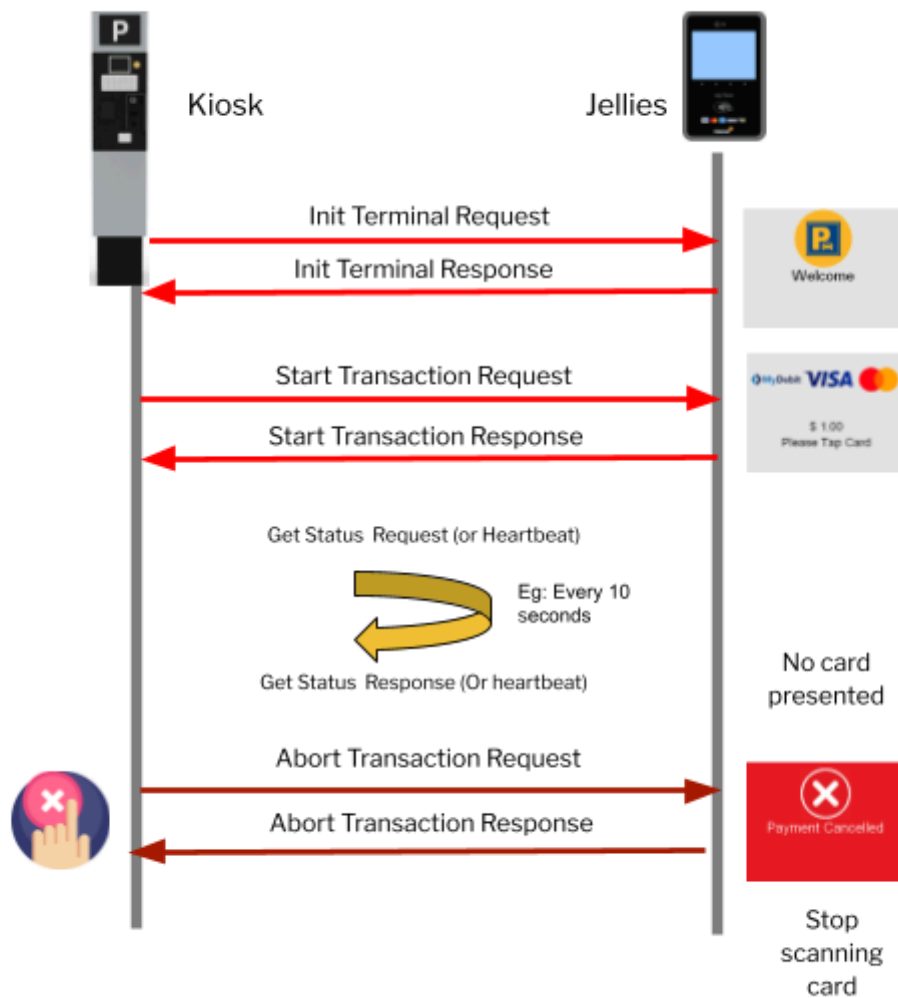


5.4 Parking Exit



5.5 Abort Transaction

This flow is applicable to the “Cancel” scenario for example: 1) customer cancels buying from vending machine, 2) customer removes the car from loop sensor.



Appendix A

Error Code

Error Code	Description
0000	Successful - Command executed successfully / Transaction Completed
10XX	Failed - Transaction Declined by Bank Host XX - Response Code from bank host 1005 - Do not honor 1051 - Insufficient Fund
1101	Failed - Transaction Failed
1102	Failed - Transaction Error
1199	Failed - Generic Error
2001	Failed - Invalid Data
2003	Failed - Invalid Condition
3001	Failed - Card Invalid
3002	Failed - Card Blacklisted
3003	Failed - Card Not Allow
3004	Failed - Card Rejected by host
3005	Failed - Multiple Card
3006	Failed - Card Expired
3007	Failed - Insufficient Fund
3008	Failed - Entry Not Clear
3009	Failed - Entry Not Found
3101	Failed - Bank Connect Error
3102	Failed - Bank Timeout
3201	Failed - Aborted by Kiosk
9001	Failed - Terminal Busy
9101	Failed - Terminal Not Active Controller has to resend initTerminal in order to activate the reader.
9102	Failed - Terminal Active
9103	Failed - LUS Disconnected
9999	Failed - General Error

Reader State

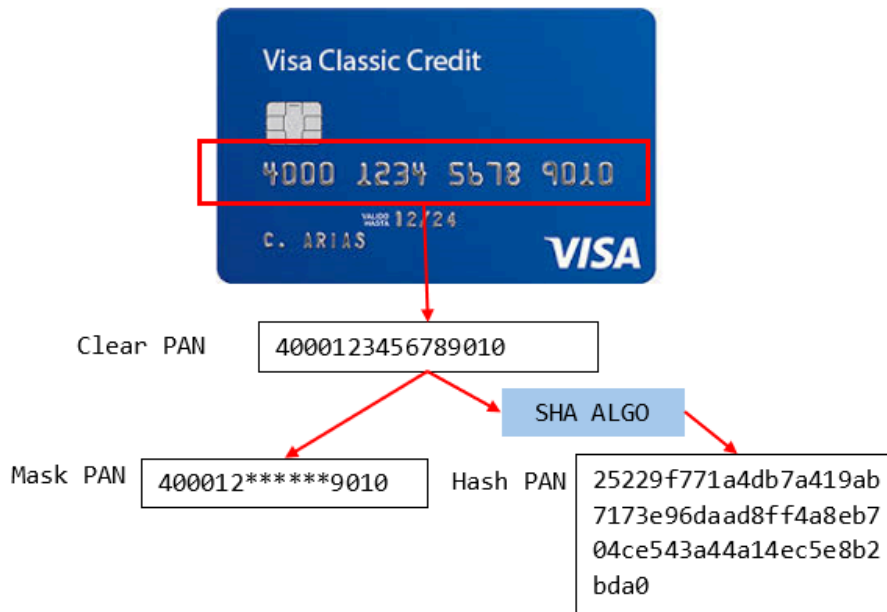
State	Description
10	Idle
20	Card Scanning
22	Card Detected
23	Card Scanning Error
30	Host Authorizing
31	Host Authorizing Failed
32	Host Authorizing Connect Failed
33	Host Authorizing Timeout
40	Completion
36	Card Scanning - Parking Entry Check
37	Card Verification- Parking Entry Check
38	Card Detected - Parking Entry Check
39	Card Scanning - Parking Exit
3A	Card Authorizing - Parking Exit
3B	Card Detected - Parking Exit
3C	Completion - Parking Enxit
41	Card Scanning - Parking Entry
42	Card Authorizing - Parking Entry
43	Card Detected - Parking Entry
44	Completion - Parking Entry
60	Finish Txn
F0	Rebooting

Scheme List

State	Description
01	VISA
02	MasterCard
03	MyDebit
11	Tng
12	Tng Abt
13	Ezlink
14	Ezlink Abt
15	Nets

Appendix B

Mask Pan and Hash Pan



Example

Clear PAN	Mask PAN	Hash PAN (20bytes, 40chars)
4000123456789010	400012*****9010	25229f771a4db7a419ab7173e9 6daad8ff4a8eb704ce543a44a1 4ec5e8b2bda0
4000127654329010	400012*****9010	3521122e5308163881cd1d31dff 6be11f6b8dbc9fee8083d93a523 6761f85e62

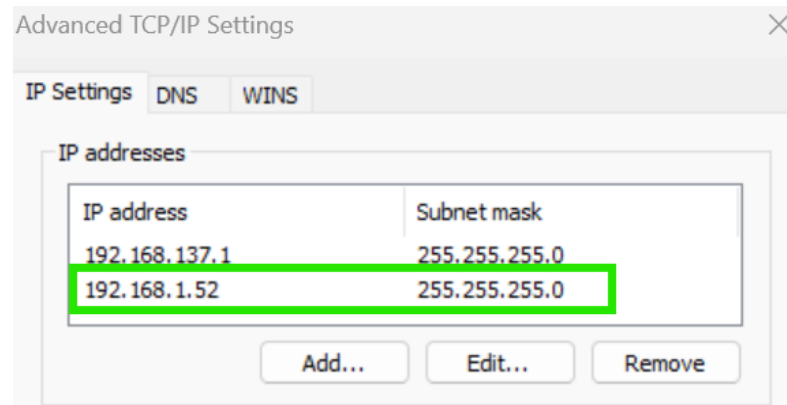
Use Mask PAN for report – Human Recognizable.

Use Hash PAN for system matching. Prevent duplicable Mask PAN.

Appendix C

Set reader IP using TCPI tool

1. Ensure to configure an additional IP at IPV4 of same subnet with the reader 192.168.1.x. (Eg: 192.168.1.52) and ping to test the connection with reader.



2. Open TCPI tool

☐ Name	Date modified	Type	Size
☒ tcpiSim+-20240716	10/28/2024 9:47 AM	File folder	

3. Launch TCPI

☐ Name	Date modified	Type	Size
Newtonsoft.Json.dll	10/28/2024 9:47 AM	DLL File	507 KB
Newtonsoft.Json.xml	10/28/2024 9:47 AM	Microsoft Edge HTM...	502 KB
☐ tcsimulator.exe	10/28/2024 9:47 AM	Application	104 KB
tcsimulator.exe.config	10/28/2024 9:47 AM	VisualStudio.config.1...	1 KB
tcsimulator.pdb	10/28/2024 9:47 AM	Program Debug Data...	158 KB
tcsimulator.vshost.exe	10/28/2024 9:47 AM	Application	23 KB
tcsimulator.vshost.exe.config	10/28/2024 9:47 AM	VisualStudio.config.1...	1 KB
tcsimulator.vshost.exe.manifest	10/28/2024 9:47 AM	MANIFEST File	1 KB

- Enter the current reader's IP address, and click "connect". Once connected, change the operator mode to "2" (offline mode) and click "Req".

Coherent TCS+ V1.0.0.0

TCS TCS2 Perso Util Parking

IP: 192.168.1.199
Port: 5000
Connect
Disconnect

TCPI JSON
Header Info
Plaza ID: P01 Lane ID: L01 Api Ver: 1.0 Operation Mode: 2 Lane Type: 3 Signature Key:
Init Terminal
Double Click for DT: 2024-11-18 10:00:09 Title: Selamat Datang Page: 1 Req
Get Status
Double Click for DT: 2024-11-18 10:00:09 Req
Init Card
Double Click for DT: 2024-11-18 10:00:09 Fare Class: 1 Title: Sentuhkan Kad Message: Trigger Req
Init Txn
Double Click for DT: 2024-11-18 10:00:09 Txn Type: 1 Fare Amount: 350 Fare Class: 1 Title: Tap Entry Dt: Vehicle No: Trigger Req
Card Detected
Double Click for DT: 2024-11-18 10:00:09 Card Read
Double Click for DT: 2024-11-18 10:00:09
Capture Txn
Double Click for DT: 2024-11-18 10:00:09 Txn ID: TXNMS2023052500000000001 Fare Amount: 350 Title: Terima Kasih Message: Jumpa Lagi Req
Abort (Cancel Card)

General Info
Trace ID: 20241118100009
Next Loop Second: 2
Auto Proceed
Manual Message
Auto Click Dt
Join
Header Type

- Select "perso" tab and type in new reader IP and "save changes".

Coherent TCS+ V1.0.0.0

TCS TCS2 Perso Util Parking

IP: 192.168.1.199
Port: 5000
Connect
Disconnect

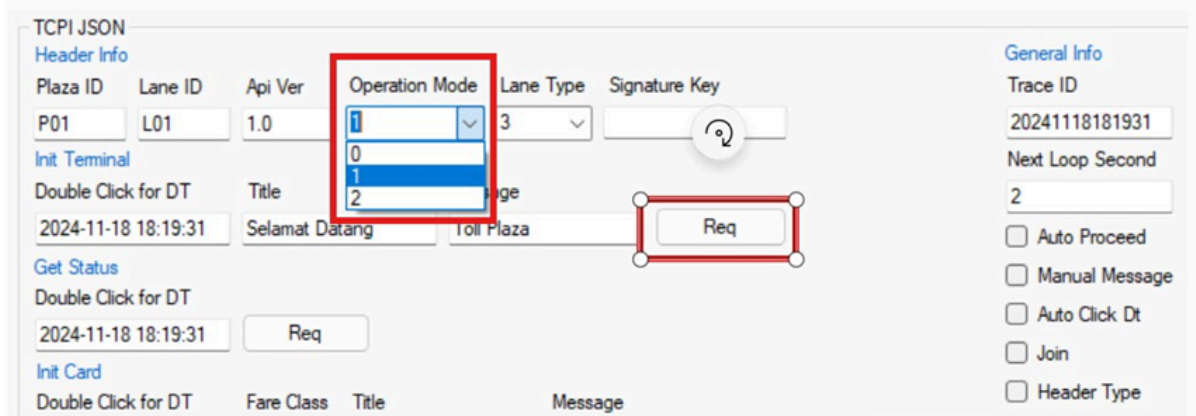
Reader IP Address
IP Address: 192.168.1.50 Static DHCP
Netmask: 255.255.255.0
Gateway: 192.168.1.1
Save Changes
*Changes require reboot (Util Page) in order to take effect

LBS
IP Address: 192.168.1.48
Port: 7077
Save Changes
*Changes require reboot (Util Page) in order to take effect

- Go to "Util" tab and click "reboot".



7. Init terminal in operation mode “1” (Live mode).



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